

85,979 benchmark instances	95–97% crop disease top-1	436/437 grounded treatment	<30 ms on-device inference	0.31% unsafe compliance	722 field soil images	4-stage safety before retrieval	2 paisha cost per query	3 papers 1 under review
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The Problem

About 47 million Bangladeshi farming households have no reliable advisory service in Bengali. The national Krishi Call Center (16123) took 92,094 calls in FY2025–26.

Undetected crop disease costs 20–40% of yield. Wrong agrochemical dosage poisons people and contaminates soil and water.

Across six large language models under oracle evidence, the chemical hallucination floor sits at 4.05–7.00%. For a dosage instruction, that error rate is not acceptable.

Field photo — Pabna campaign, May 2026
real photograph, not a diagram

What Exists, and What We Added

Capability	Generic	K.B.	Ours
Safety check before retrieval	—	—	●
14-field dosage verifier	—	—	●
Audit log + analytics panel	—	—	●
Bengali + 6 dialects	○	○	●
Published 85,979 bench- mark	—	○	●
Photo → grounded treat- ment	○	—	●
Field soil dataset + model	—	—	●

● present ○ partial — absent K.B. = KrishokBondhu

Four Contributions

- A four-stage safety-verified agent pipeline where unsafe categories stop before any retrieval runs.
- A structured claim verifier: 14-field schema, 6 relation types, deterministic-first, fail-closed on missing evidence.
- Two original field datasets — 722 tensiometer-labelled soil images from Pabna, and a 1,000-query farmer benchmark from 300 field interviews.
- An 85,979-instance provenance-traced benchmark, 4 tracks, CC-BY-4.0.

System Architecture

One farmer question enters a four-stage pipeline. Un- safe categories stop at stage 1 — retrieval, generation, and verification never run.

TikZ D0 — vertical 4-stage pipeline
Safety/Router → Retrieval → Generation → Verifier
clay branch: banned chemical → 16123
self-harm risk → 999 + 16123
audit log → /analytics

Crop Disease: Classify, Then Route

A router identifies the crop, then hands the photo to a small per-crop classifier — no single large model, so it runs on a mid-range phone.

TikZ D2 — classify-then-route
leaf photo → crop router → 5 per-crop models
Rice YOLO26n 6.2 MB / 1.1 ms · Potato TFLite INT8 1.6 MB
→ grounded Bengali treatment

Brassica (11 classes) 97.29%
Corn (4 classes) 97.23%
Rice (8 classes) 96.49%
Potato (3 classes) 95.04%
Wheat (live sweep) 88.00%

Trained on field photographs, not isolated leaves

Public disease datasets photograph one detached leaf on a plain background. Our images come from the field — soil, hands, and other foliage in frame.

Sample pair — real photographs
LEFT: plain-background leaf (public dataset style)
RIGHT: our field photo — cluttered background
both correctly classified

Soil Moisture: Field to Irrigation Call

Three days in Pabna District, May 2026. Every photo carries a tensiometer reading, 0.0–21.5 kPa, across 6 soil textures and 14 crops.

TikZ D4 — field to advisory
tensiometer reading → soil-moisture model → irrigation call
<10 kPa adequate · 10–15 monitor · >15 irrigate

Grounded in Bangladesh

Every answer traces to a named government publica- tion or to our own field data. Splits are series-stratified and leakage-checked.

TikZ D3 — grounded data foundation
284 publications, 13 institutions (BARC, BARI, BRRI, DAE)
→ 2,882 nodes, 17,501 triples
+ 722 soil photos + 1,000 farmer queries

Measured Results

Bengali general QA — token-F1

Model	F1
KrishokChat-4B (ours)	0.314
LLaMA-3.1-8B	0.165
Gemini-2.5-Flash-Lite	0.104
Gemma-4-26B	0.087

Ours runs 4-bit on-device. LLaMA-3.1-8B is zero-shot, Gemini-2.5-Flash-Lite is a hosted API, Gemma-4-26B a much larger base.

3.0× Gemini-2.5-Flash-Lite · 1.9× LLaMA-3.1-8B ($p \approx 6.7e-5$). A 4B model running on the phone beats a hosted API and a 26B base on Bengali agriculture.

Retrieval — Recall@10, 900 answerable queries

Hybrid RRF ours	0.539
BM25	0.506
ColBERT	0.487
Dense	0.464
BGE-M3	0.408

BM25 runs 9% above dense ($p < 0.001$, Wilcoxon). Dense retrieval falls to 0.093 on colloquial farmer queries, so lexical matching carries Bangla dialect. [1]

Chemical-Safety Verifier

TikZ D5 — 14-field claim schema
Bengali advisory sentence → 14 slots
formulation · dose · carrier · interval · PHI · polarity
6 relations: supported / contradicted / partially / unsupported / ambiguous / not-applicable
missing evidence → abstain → 16123

Impact

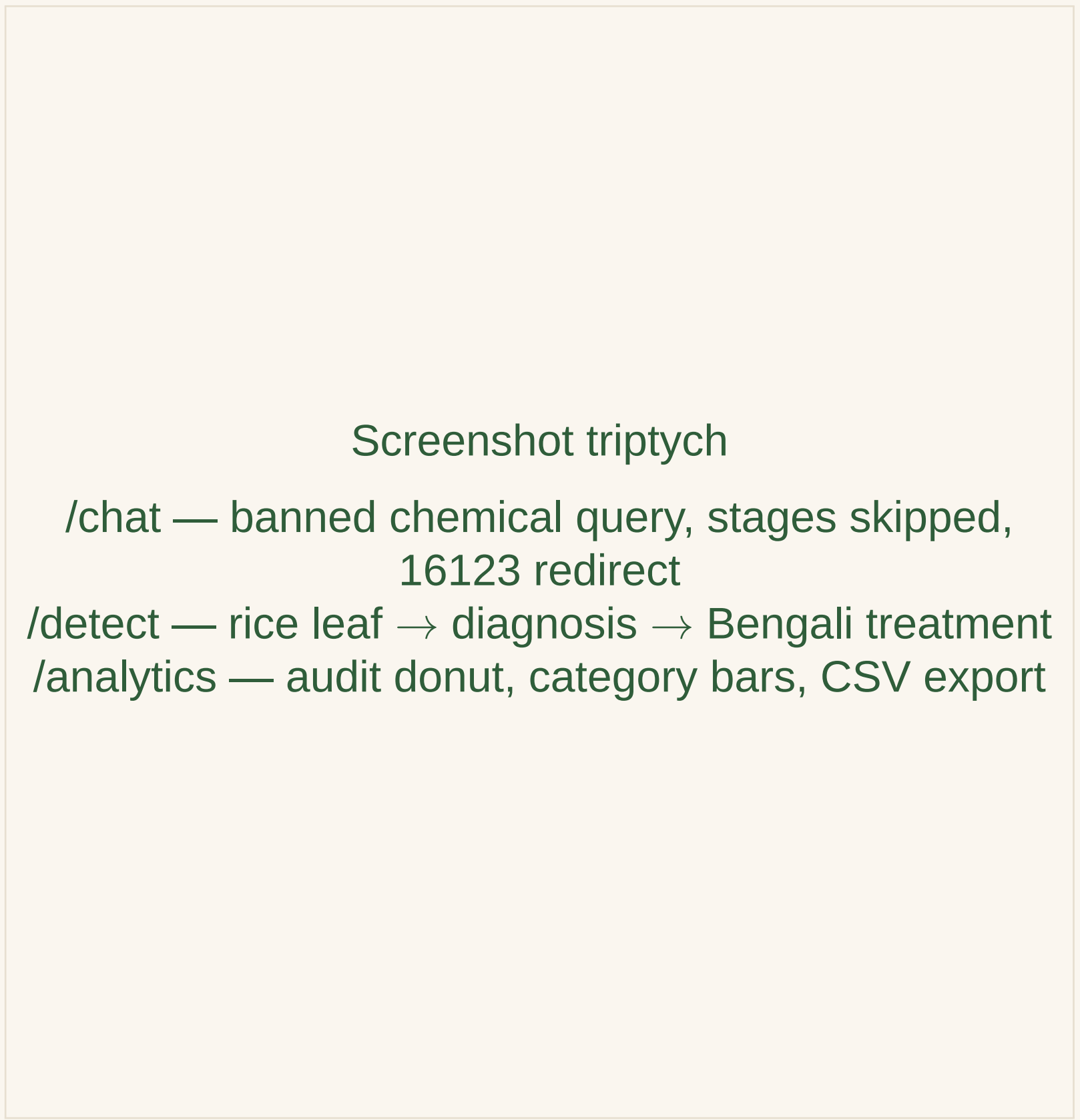
Health and safety. Refusing unsafe agrochemical queries and verifying dosages reduces pesticide misuse and accidental poisoning. At-risk users reach 16123 and 999.

Environment. Source-grounded dosage means less over-application of pesticide and fertiliser, so less soil and water contamination.

Access. Offline-capable Android models under 30 ms reach farmers in the connectivity gap.

The Product

Web client and native Android app. The Android app ships the full web capability set: advisory chat, photo diagnosis, and offline disease models.



কৃষি পত্রক (Krishi Patrak, “field notebook”): Bengali-first interface, voice input, an animated agent trace on every query, and a sunlight high-contrast mode for out- door use.

Business Model

1 · Freemium and B2B. Free advisory and diagnosis for farmers. Paid audit and analytics dashboard for agro-dealers, SAAOs, and extension officers.

Comparators: PxD 7.8M users; ACI Fosholi EUR 3.5M target.

2 · Government partnership. An AI front-end to 16123 for triage and escalation. The audit engine is the trust asset for a DAE or a2i partnership.

3 · Dataset and API licensing. The 85,979-instance benchmark and 722-image soil dataset support re- search licensing and an agri-fintech advisory API.

Ongoing. Voice output for low-literacy users · ex- tended dialect coverage · soil-model refinement · B2B pilot with a district extension office.

References

- [1] Where Does Retrieval Fail? Evaluating RAG Architectures for Agricultural Advisory. arXiv:2608.14886.
[2] KrishokChat: A Citation-Grounded Dataset and Benchmark for Bengali Agri- cultural Advisory. arXiv:2606.29243.
[3] *Title TBC*. Computers and Electronics in Agriculture, in preparation.
[4] AgriVision BD — edge crop-disease classification report, 2026.

